Customer economics lab

Are we moving in the right direction as a company?

Connect separate CSV exports, inspect customer cohorts, and distinguish predicted value from incremental marketing effects.

CRM	Renewals
Transactions	Product events
Feedback	Campaign spend
Source credit	Assignments

How to read the slide colors

Concept	Definitions and distinctions to use when interpreting the data.
Case application	The fictional subscription business and its connected records.
Evidence interpretation	Observed results, predictions, and what each can establish.
Group decision	One focused judgment, discussed with your group.

Three ideas guide the entire lab

1. Connected customer evidence

A Universal Customer View connects identifiers, relationships, engagement and perceptions without confusing their meanings.

2. Customer economics over time

Joining cohorts distinguish customer age from calendar time. Retention, contribution and acquisition cost determine whether relationships create value.

3. Incremental marketing value

A randomized campaign comparison asks what changed because of the campaign. A high customer LTV alone cannot answer that question.

AI proposes the work. Saved SQL, data checks, held-out outcomes and your judgment make the work reviewable.

Eight exports describe different objects

Source exports	One row represents	Question it helps answer
People	A person, including nonbuyers	Who is eligible and what need was stated?
Renewals / Transactions	A subscription period / a financial event	Who remains, who paid, and what was refunded?
Product events / Feedback	An interaction / a response	What happened and what was reported?
Spend / Source credit	A campaign date / an acquired person	What did acquisition cost, and who received credit?
Assignments	An eligible person in an experiment	What was assigned, including for nonbuyers?

Ask AI to inspect the files, identify their grains, and draw the system before it joins them.

Group decision 1: who needs which onboarding route?

Product can offer two routes: a fast first result or guided practice. Which existing grouping should inform this design?

Grouping	Two recorded groups	What the field describes
Age band	18–34 / 35+	A demographic characteristic.
Stated need	Save time / Build skill	The goal stated before acquisition.

Choose one grouping and explain why it is relevant to this onboarding objective.

Use the conceptual journey maps to propose the routes. Later, use the SQL results to test whether the grouping explains value.

Turn the connected view into a reproducible query

Inspect raw views

Keep source columns and row meanings.

Define the customer clock

First paid month sets cohort age index 0.

Build SQL outputs

Save the exact SQL and ordered results.

Reconcile the totals

No duplicate money; no missing controls.

Prompt to try

“Show the SQL that calculates retained customers and contribution by joining month and customer age. Explain the denominator and preserve unobserved future ages as missing.”

A SQL script makes a calculation inspectable; the data definitions determine whether it answers the right question.

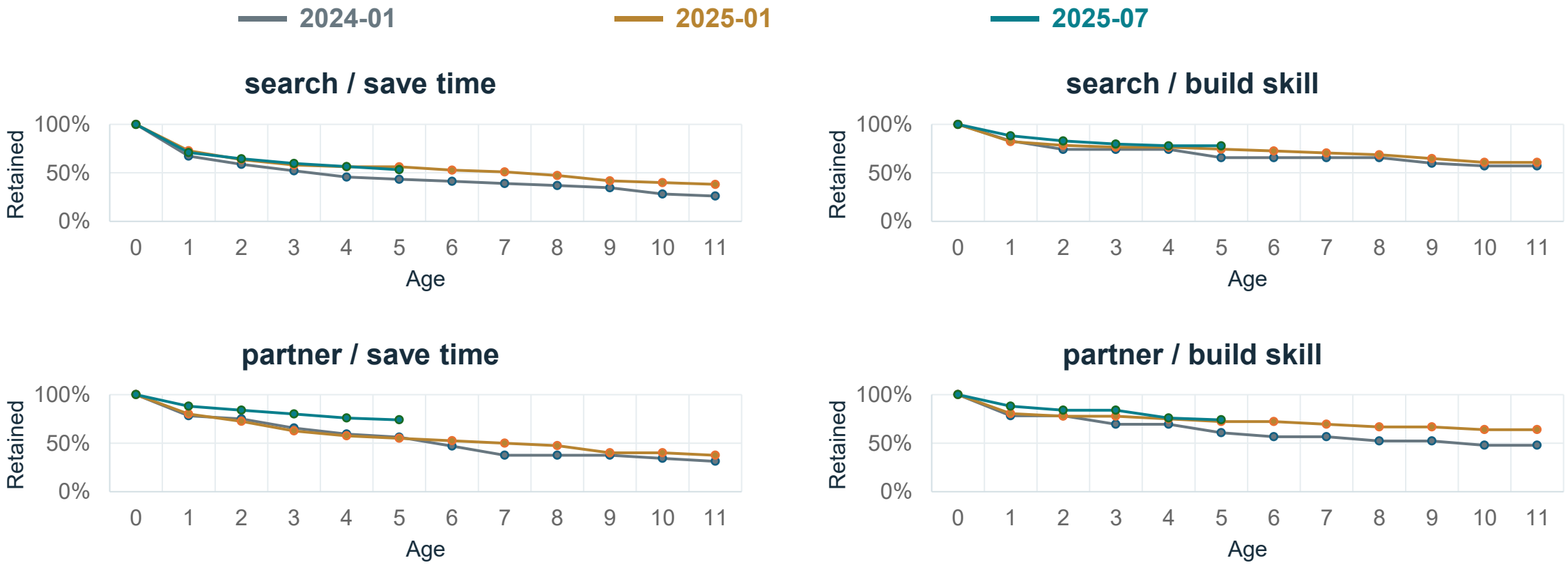
A cohort follows the same original customers

Joining cohort	Age 0	Age 1	Age 2	Age 3	Age 4	Age 5
2024-01	100%	76%	70%	64%	60%	55%
2024-10	100%	82%	74%	71%	70%	67%
2025-07	100%	83%	78%	75%	71%	69%
2025-12	100%	—	—	—	—	—

Across: follow one joining cohort. Down: compare different cohorts at the same customer age.

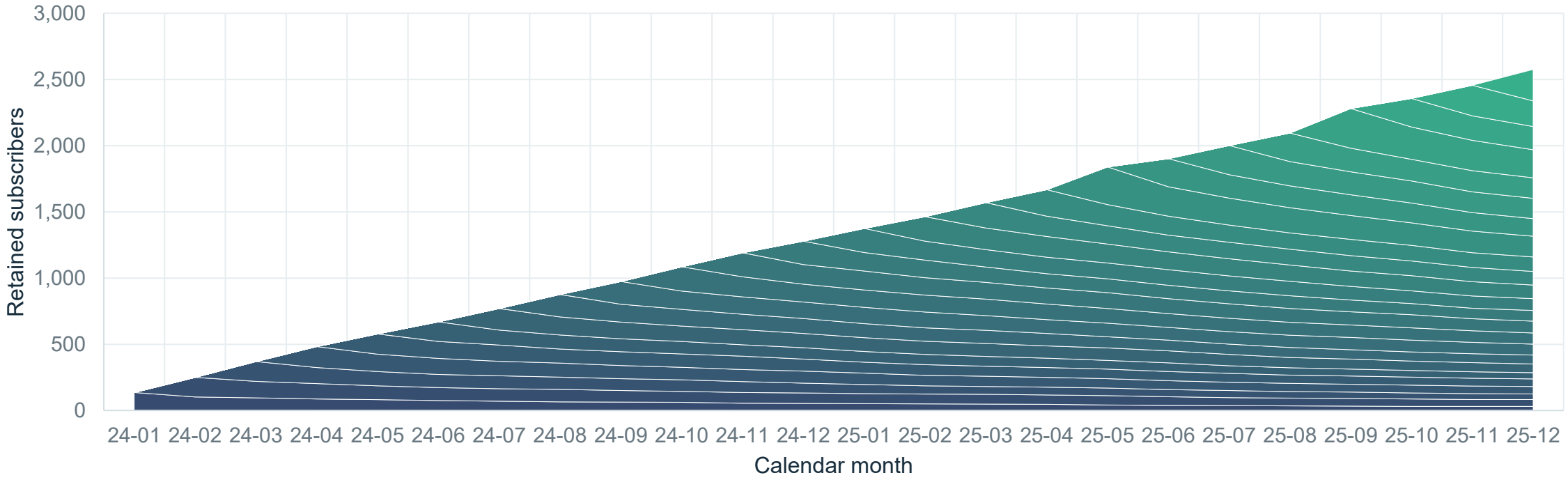
A blank is not zero. The newest cohort has not had time to reach later ages.

Group decision 2: is improvement visible within groups?



Compare joining cohorts inside the same channel and need. What improves at matching ages—and what remains unobserved?

Joining cohorts accumulate into the company



Each band is one joining cohort. The height is the retained customer base, not cumulative acquisitions.

Group decision 3: which campaign merits a repeat pilot?

Each campaign randomized 250 eligible prospects to treatment and 250 to its own no-campaign control.

Test	Incremental contribution	Additional media	After media / eligible prospect
search / save time	$658.80	$650.00	$0.04
partner / build skill	$1,161.65	$900.00	$1.05

Which tested campaign would you prioritize for a repeat pilot, and what limits that recommendation?

Keep each test's control and all nonbuyers. These point estimates do not establish an optimal marginal budget allocation.

LTV extends the observed history under assumptions

Observed months

Use the contribution already recorded for the cohort, including zeros for customers who left.

Unobserved months

Predict retention and spending for the remaining months of one common 12-month horizon.

Per acquired customer

Discount each monthly contribution at 1% per month, divide by original cohort size, then subtract CAC for net LTV.

Model check: forecast outcomes that were withheld, and compare the error with a simpler baseline.

Cash payback is undiscounted. A model-based LTV prediction is not a causal return on marketing spending.

Resolve the opening question with a stated next action

Are we moving in the right direction as a company?

Evidence	What belongs in your answer
Observed customer economics	Same-age retention, revenue, contribution, CAC and cash payback.
Company dynamics	Whether retained older cohorts support the growing active base.
Predicted value	A common horizon, held-out accuracy and explicit forecast assumptions.
Experimental effects	The tested campaign’s contribution after media relative to its control.

Record your verdict and recommended action. The LaTeX report preserves both, together with the evidence and saved SQL.